



FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

INTEGRATIVE PROGRAMMING AND TECHNOLOGIES III

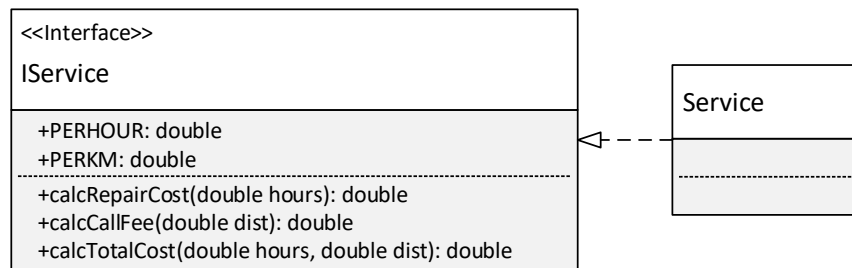
TEST 3

DATE:	23 May 2024
TIME:	40 minutes
MARKS:	28
EXAMINER:	BOAMAH-ABU, CHARLES (PhD)
NO. OF PAGES:	3 (Including cover page)

INSTRUCTIONS/REQUIREMENTS

1. This examination is closed book
2. Answer all questions
3. Where stated "briefly", please give a short answer
4. Ensure your handwriting is readable

NLM Locksmith offers call-out service to clients. You are to write an RMI application which calculates the total fee for a call-out service. The UML diagram below shows the remote interface and remote class of the application.



The interface contains:

- `calcServiceFee(double hours)`: takes the number of **hours** worked as parameter and returns the service fee at R200 per hour
- `calcCallFee(double dist)`: takes the **distance** to the client as parameter and returns the call-out fee at R50 per kilometre
- `calcTotal(double hours, double dist)`: takes the **hours** and **distance** as parameters, calls the appropriate methods, and returns the total fee (i.e., the service fee and call out fee)
- `PERHOUR` is a constant, the rate per hour (i.e., R200)
- `PERKM` is a constant, the rate per kilometre (i.e., R50)

Use the information above to answer **Questions 1 to 5** below.

1. Write the code for `IService`. (11 marks)
2. Write the class header for `Service`. (3 marks)
3. Write the code for the following methods in `Service`.
 - a. Empty constructor. (2 marks)
 - b. `calcTotal(double hours, double dist)`. (4 marks)
4. Complete the code segment below for a class `Server` which creates, registers and binds an instance of `Service` (called `s`) with the reference “`service`”. Assume the class `Service` has been implemented. Use port 2024. (5 marks)

```

...
...
public class Server {
    public static void main(String[] args) throws Exception{
        _____
        _____
        _____
    }
}

```

5. Complete the code segment below for a class **Client** which creates an instance of the remote object (called **is**), calls a method of **is** (with hours=2hours and distance=2km) and displays the total fee. Assume IService has been implemented, the server is "localhost" and the port is 2024. (5 marks)

```
...
...
public class Client {
    public static void main(String[] args) throws Exception{
        _____ is= _____
        double hours=2;
        double dist=2;
        System.out.println("Hours="+hours);
        System.out.println("Distance(km)="+dist);
        System.out.println("Total fee=R"+ _____);
    }
}
```